firms affiliated with a self-aggregator,[1091] and another stated that requiring each affiliated entity to aggregate and build its own market data systems would be a needless drain of resources.[1092] This commenter further stated that self-aggregators should be permitted to share self-aggregated data with their affiliates because a market maker should be able to know when facilitating interest for an agency affiliate that its view of the quoted market is consistent with that of the affiliate.[1093] Another commenter suggested that the Commission allow self-aggregators to use consolidated market data in handling and routing orders on behalf of the broker-dealer's customers, including in cases where customer business is conducted through an affiliate, without being required to pay separate fees for that purpose.[1094] However, one commenter stated that permitting a self-aggregator to disseminate consolidated market data to its affiliates would allow

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1091 See MFA Letter at 5.

1092 See Susquehanna Letter at 5. In addition, the commenter argued that "self-aggregator organizations should not be faced with the disruptive and needlessly costly and burdensome choice of (1) developing and maintaining redundant consolidated data sets for each respective user within the organization, (2) registering as a CC and assum[ing] the related obligations and liabilities even though it never wanted to be in that business, or (3) subscribing to the outside services of registered CCs (again on a redundant basis for each entity within the organization), whose quality and/or cost efficiency may be less, and over whom such organization would have less control to customize or improve services, or to remediate problems." Id. at 6.

1093 See Susquehanna Letter at 4.

1094 See IEX Letter at 9. One commenter expressed the view that sharing consolidated market data within a single affiliated entity organization, under common beneficial ownership and senior hierarchical management, is not performing the functions of a competing consolidator because the consolidated market data is not intended for public dissemination in connection with commercial competition of exchange data feeds. See Susquehanna Letter at 5–6.

the self-aggregator to perform the function of a competing consolidator without the burdens of being a competing consolidator.[1095]

The Commission believes that self-aggregators should be permitted, as an internal use, to make consolidated market data available to their affiliates that are registered with the Commission. A broker-dealer or RIA that is affiliated with a self-aggregator may require consolidated market data to fulfill its regulatory obligations, as described above. In addition, as noted above, the Commission has the authority to examine the registered affiliated entities of a self-aggregator and would be able to determine how the self-aggregator provides consolidated market data to a registered affiliate and how the registered affiliate uses that data. Therefore, a self-aggregator will be permitted to share consolidated market data only with affiliates that are registered with the Commission.

An affiliate of a self-aggregator that is not registered with the Commission, however, may not have the same regulatory obligations as registered entities,[1096] and the Commission does not have the authority to examine a self-aggregator's unregistered affiliates. In addition, as discussed above, the Commission believes the widespread dissemination of consolidated market data must be subject to Commission oversight and, accordingly, must be performed by

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1095 See Data Boiler Letter I at 60.

1096 For example, broker-dealers execute customer orders and must comply with Regulation NMS related to the routing and execution of orders in the national market system, including Rules 611 and 610(d). In addition, broker-dealers are subject to the duty of best execution. RIAs and SROs also have regulatory obligations, as discussed above in Sections III.D.2(a)(ii) and III.D.2(a)(i), respectively.

competing consolidators. Competing consolidators will be subject to the registration, disclosure, and other regulatory requirements in Rule 614 and Form CC.[1097]

(ii) Sharing Consolidated Market Data with Customers

Several commenters stated that broker-dealers that self-aggregate should be permitted to display their self-aggregated data to their customers without registering as a competing consolidator or becoming a Regulation SCI entity.[1098] One commenter stated that if brokers are not permitted to share consolidated market data with their customers, proprietary traders and high frequency firms would add to their significant data cost advantage over retail investors and the two-tiered data system would be preserved.[1099] The commenter further stated that the Commission should allow self-aggregators to display consolidated market data to their customers

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1097 See supra Section III.C.7(a)(iv).

1098 See SIFMA Letter at 12 (stating that broker-dealers that self-aggregate should be permitted to display their data to their customers, subject to the requirements of the Vendor Display Rule, without being required to register as a competing consolidator or Regulation SCI entity); TD Ameritrade Letter at 12 (stating that registered broker-dealers should be allowed to use self-aggregated consolidated market data for display to their brokerage clients, without further sale or redistribution to unaffiliated third parties; the proposal would require a broker-dealer self-aggregator that wishes to provide its self-aggregated data to its clients to invest time and resources into becoming a competing consolidator compliant with Regulation SCI requirements, or to buy consolidated market data from competing consolidators for display purposes); Schwab Letter at 2, 6–7 (stating that self-aggregators should be allowed to share consolidated data with their customers on a not-for-profit and non-redistribution basis, but not with external parties, and should not be required to comply with Regulation SCI because they are not holding themselves out as a "public utility").

1099 See Schwab Letter at 7. See also TD Ameritrade Letter at 11–12 (stating that the internal-use limitation on self-aggregated data could disadvantage retail investors because a broker-dealer would be compelled to purchase consolidated data from a competing consolidator, and those able to pay the competing consolidator for faster speeds could "get to the market" more quickly).

to encourage competition among the competing consolidators, enable retail investor access to data with the least amount of latency without additional cost, and allow broker-dealers to share with their customers the same view of the same core data.[1100] Another commenter stated that registered broker-dealers should be allowed to share self-aggregated consolidated market data with their brokerage clients without registering as competing consolidators, noting that the benefits of Regulation SCI compliance are "inherent in the registered broker-dealer regulatory regime for continuity of operations and display of the data."[1101]

Under the amendments, self-aggregators will not be permitted to disseminate or otherwise share or make available consolidated market data to any persons, including their customers or clients.[1102] The dissemination of consolidated market data entails a different process from self-aggregating consolidated market data for internal uses (e.g., for order handling, routing, and execution). Self-aggregators are not subject to the regulatory regime established for competing consolidators, which is designed to ensure that consolidated market data is reliable,

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1100 See Schwab Letter at 6.

1101 TD Ameritrade Letter at 12. The commenter noted that broker-dealers are subject to FINRA Rule 4370 (establishing requirements for designing business continuity plans which require data backup and recovery, mission critical systems, and alternate location requirements, among others) and FINRA Rule 4380 (requiring mandatory participation in FINRA business continuity and disaster recovery ("BC/DR") Testing under Regulation SCI if determined necessary by FINRA). See id. at n. 36.

1102 The Commission has revised the definition of self-aggregator to further clarify that a self-aggregator may not disseminate or otherwise make available consolidated market data, or components of consolidated market data, as provided in §242.600(b)(20), to any person other than an affiliate that is registered with the Commission.

resilient, and accurate. The Commission believes that entities that deliver consolidated market data to third parties should be subject to such standards.[1103]

The Commission believes that investors and other non-registered entities should receive consolidated market data from entities that are subject to a regulatory regime that is designed to ensure the data they receive is reliable, resilient, and accurate and that they are able to assess such reliability, resiliency, and accuracy on an ongoing basis. Self-aggregators are not subject to such standards or requirements and therefore will not be permitted to disseminate or otherwise make available self-aggregated consolidated market data with customers, clients, or non-registered affiliates.

(c) Self-Aggregators and Market Data Fees

One commenter stated that exchanges seeking the business of self-aggregators might offer "enterprise license" pricing packages that would allow a firm and all of its affiliates to receive proprietary data for one price, effectively allowing the self-aggregator to share data with its affiliates.[1104] An exchange seeking to establish "enterprise license" pricing packages for

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[1103] Competing consolidators will be registered with the Commission and will be subject to systems integrity and operational capability standards that will help to ensure the accuracy and availability of the consolidated market data that they produce. See infra Section III.F. Competing consolidators also will have certain responsibilities and obligations, including obligations to disclose publicly operational information and performance metrics, which will help to ensure transparency, accountability, and oversight, and obligations to ensure the integrity, quality, and resiliency of consolidated market data. See supra Section III.C.8. Self-aggregators, by contrast, will not be subject to similar requirements in the collection, consolidation, or generation of consolidated market data because they will not disseminate consolidated market data or otherwise make consolidated market data available to persons other than affiliates registered with the Commission.

[1104] Nasdaq Letter IV at 57, n. 80.

proprietary data would be required to file those proposed fees with the Commission pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder, and such fees must satisfy the statutory standards of being an equitable allocation of reasonable fees, dues, and other charges,[1105] not being unfairly discriminatory,[1106] and not an undue burden on competition.[1107]

(d) Two-Tiered Market and Potential Advantages of Self-Aggregators

The Commission requested comment on the potential latency advantage of self-aggregators.[1108] One commenter stated that self-aggregators' latency advantage would not be material.[1109] In contrast, another commenter stated that the latency advantage would not be minor, given the time increments currently used in the market and the likelihood of finer increments over time.[1110] The commenter questioned whether the Commission had considered eliminating the self-aggregator category and requiring all market participants to receive data from one or more competing consolidators, or requiring SROs to delay the provision of data to match the latencies introduced by competing consolidators.[1111] One commenter stated that,

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1105 Section 6(b)(4) of the Exchange Act.

1106 Section 6(b)(5) of the Exchange Act. See also Rule 603 of Regulation NMS.

1107 Section 6(b)(8) of the Exchange Act.

1108 See Proposing Release, 85 FR at 16791.

1109 See Clearpool Letter at 10.

1110 See FINRA Letter at 8. See also Data Boiler Letter I at 59 (stating that the latency advantage would be material).

1111 See FINRA Letter at 8–9. See also Angel Letter at 8 (suggesting that the Commission embargo the exchanges from releasing any data until the consolidators have had sufficient time to process the data to create a more level playing field); Healthy Markets Letter at 3 (suggesting that the Commission remove the latency advantage of exchange proprietary data feeds by requiring all market participants to receive data from SIP distributors).

because of "the additional inherent latency in third-party aggregation," it is unlikely that broker-dealer algorithms would be competitive without self-aggregation.[1112] Another commenter stated that the proposal would create a tiered market in which broker-dealers have systematically better and more timely access to market data than registered investment advisers and noted that self-aggregators would have both a speed and potential cost advantage over those who receive consolidated market data from competing consolidators.[1113] Other commenters similarly argued that the proposal would create a two-tiered market data system comprising self-aggregators and those who receive data from competing consolidators.[1114]

The Commission acknowledges that, unlike self-aggregators, competing consolidators would need to transmit consolidated market data to their customers, but does not believe that this would lead to the development of a two-tiered market. Latency sensitive customers of competing consolidators are likely to be co-located in the same data centers as their competing consolidators, so the transmission time between the servers of the competing consolidator and its customer will be exceedingly small. In many cases, self-aggregators may be located in the same data centers, and the potential latency differential between a self-aggregator and competing

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1112 See NBIM Letter at 4.

1113 The commenter stated that self-aggregators would be able to receive the data necessary to generate consolidated market data at the price established by the effective national market system plan(s), while market participants that receive consolidated market data from competing consolidators might have to pay a premium over that amount to compensate the competing consolidator for its services. See MFA Letter at 4.

1114 See, e.g., FINRA Letter at 8; NYSE Letter II at 23 (stating that the proposal would continue the two-tiered structure, with participants that can afford to act as self-aggregators able to obtain and use that data faster than those relying on competing consolidators); STANY Letter II at 6 (stating that the proposal would replace the existing two-tiered structure between SIPs and proprietary data feeds with, at minimum, a different two-tiered structure between self-aggregators and competing consolidators); Nasdaq Letter IV at 3 (stating that self-aggregation would add market-wide disparities in terms of data content and speed).

consolidator resulting from the extra hop that competing consolidators add to the process of data consolidation and dissemination could amount only to the period of time it takes to send a message from one server (i.e., a competing consolidator's server) that is located in close proximity to another server (i.e., a subscriber's server) and connected via a cross connect.

The Commission expects that market participants that elect to aggregate consolidated market data, whether competing consolidators or self-aggregators, will innovate and compete aggressively on the efficiency and cost-effectiveness of their aggregation technologies. The Commission believes that the development and implementation of the technology to collect, consolidate, and generate consolidated market data will create opportunities for latency efficiencies that are of substantially greater magnitude than the transmission time between the server of a competing consolidator and its customer. Competing consolidators, for example, may benefit from economies of scale that allow them to offer a very low-latency product more cost effectively that an individual self-aggregator. In some cases, a competing consolidator may have a latency or cost advantage, and in others a self-aggregator may have such advantages.[1115] Competition may also impact the efficiency of choices.[1116] Therefore, the Commission does not

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[1115] Self-aggregators could have a cost advantage over market participants that receive consolidated market data from a competing consolidator because self-aggregators will not be required to compensate a competing consolidator for its services. At the same time, a self-aggregator will need to have the systems capability to collect, consolidate, and generate consolidated market data, and it may use a vendor to establish connectivity to an SRO or to perform aggregation or other functions necessary for generating consolidated market data. As a result, any potential cost advantage of a self-aggregator over market participants that purchase consolidated market data from competing consolidators may not be significant.

[1116] See infra Section V.C.4(b).

believe that self-aggregators would necessarily have a systematic latency advantage over customers of competing consolidators.

(e) Fees Charged by Competing Consolidators

One commenter recommended that the Commission implement a mechanism for it to review or abrogate fees charged by competing consolidators to ensure that consolidated market data is available on terms that are fair and reasonable (i.e., reasonably related to costs) if non-broker-dealers are not permitted to act as self-aggregators.[1117] As discussed above, competing consolidator fees will be disclosed on Exhibit G to Form CC. The Commission believes that competition among competing consolidators, along with disclosure, will help to ensure that the fees charged by competing consolidators are fair and reasonable. The fees for the data content underlying consolidated market data established by the Equity Data Plan(s) will be filed under Rule 608 and must comply with statutory standards.[1118]

E. Amendment to the Effective National Market System Plan(s) for NMS Stocks under Rule 614(e)

The effective national market system plan(s) for NMS stocks will continue to play an important but modified role in the provision of consolidated market data to market participants.[1119] Today, the Equity Data Plans operate the exclusive SIPs and therefore, directly collect, consolidate, and disseminate SIP data. Under the decentralized consolidation model, the effective national market system plan(s) for NMS stocks will no longer operate the exclusive SIPs and therefore, will not be directly responsible for collecting, consolidating, and

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1117 See MFA Letter at 5.

1118 See infra Section III.E.2(c).

1119 See Governance Order, infra note 1128.

disseminating consolidated market data. The plan(s) will, however, continue to develop and oversee the national market system for consolidated market data.

1. Proposal

The Commission proposed Rule 614(e), to require the participants to the effective national market system plan(s) for NMS stocks to file an amendment to such plan(s) to reflect the decentralized consolidation model and the new role and functions of the plan(s). The Commission proposed several specific provisions to be included in the amendment, including (1) the proposed fees to be charged by the plan(s) for the data content underlying consolidated market data, (2) a requirement under the plan(s) for the application of timestamps by the SROs to the data content underlying consolidated market data, (3) a requirement under the plan for the completion of annual assessments by the plan participants of the performance of competing consolidators, and (4) a requirement for the development a list of the primary listing markets for each NMS stock. In addition, under proposed Rule 614(d)(5), the plan(s) would be required to develop the monthly performance metrics for competing consolidators. As proposed, the participants would be required to file this amendment pursuant to Rule 608 within 60 calendar days from the effective date of Rule 614.

2. Final Rule and Response to Comments

The Commission continues to believe in the importance of the use of effective national market system plan(s) in the planning, development, operation, and regulation of the national market system for the dissemination of consolidated market data. The Commission believes that joint consideration by the SROs and other market participants on the Operating Committee of such plan(s) will help to foster a consolidated market data national market system that is prompt, accurate, reliable, and fair and furthers the goal of helping to ensure that the consolidated market